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Female Sprague-Dawley rats were rendered diabetic by a single IP injection of STZ and hyperglycemia was confirmed four days after injection. Animals were examined weekly for tactile allodynia. Once allodynia was established six animals were sacrificed and plasma and dorsal root ganglia were sampled to allow for measurement of LPA upregulation. Concentrations of LPA measured in the plasma were significantly higher in the diabetic animals in comparison with non-diabetic control animals. Dorsal root ganglia of diabetic animals also showed increased staining for LPA. In a subsequent study diabetic rats were treated with either 1. a single i.v. injection (10 mg/kg) of B3 concomitant with an intra-cisternal injection (10ug/mouse); or 2) an isotype control IgG at the same dosage; or 3) vehicle. At three hours post administration of test article animals were again examined for hypersensitivity. Plasma was collected for measurement of total LPA concentration by ELISA and free concentration by KinExA. Results showed that B3 treatment increased the paw withdrawal threshold in diabetic rats in comparison with IgG control treated animals. Animals treated with B3 also showed a significant reduction in the concentration of total LPA in the plasma. When the individual animal data for paw withdrawal threshold was compared with the concentration of free LPA remaining in the plasma an inverse correlation was noted.

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Evaluation of LPA concentrations in a model of STZ-induced diabetes in rats and preliminary efficacy with an anti-LPA antibody

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1 SUMMARY

The purpose of this study was to evaluate if LPA was upregulated in a model of STZ-induced diabetes in the rat. Preliminary efficacy of an anti-LPA antibody was also examined.

Female Sprague-Dawley rats were rendered diabetic by a single IP injection of STZ and hyperglycemia was confirmed four days after injection. Animals were examined every week for the manifestation of tactile allodynia. Once allodynia was established six animals were sacrificed and plasma and dorsal root ganglia were sampled to allow for measurement of LPA upregulation in the plasma (by Mass Spectrometry) and in the DRGs by immune histochemistry using an anti-LPA antibody (LT3211). Concentrations of LPA measured in the plasma were significantly higher in the diabetic animals in comparison with non-diabetic control animals. Dorsal root ganglia of diabetic animals also showed increased staining for LPA in comparison with normal rats. The remaining animals were divided into groups and tested for mechanical hypersensitivity. These animals were then treated with either 1) a single intravenous injection (10 mg/kg) of the murine antibody B3 concomitant with an intracisternal injection (10 ug/mouse); or 2) an isotype control IgG at the same dosage; or 3) vehicle (PBS). At three hours post administration of test article animals were again examined for hypersensitivity. Plasma was collected for measurement of total LPA concentration by ELISA and free concentration by KinExA. Results showed that anti-LPA antibody treatment increased the paw withdrawal threshold in diabetic rats in comparison with IgG control treated animals. Animals treated with B3 also showed a significant reduction in the concentration of total LPA in the plasma. Levels in the plasma following treatment with B3 were similar to those in control non-diabetic animals. When the individual animal data for paw withdrawal threshold was compared with the concentration of free LPA remaining in the plasma an inverse correlation was noted.

In summary, this study demonstrated elevation of LPA in diabetic animals and that a single dose of B3 was sufficient to ameliorate the tactile hypersensitivity measured in these animals. In addition, B3 reduced the concentration of LPA in the plasma and this reduction correlated with the extent of efficacy seen.

2 INTRODUCTION

The bioactive lipid lysophosphatidic acid (LPA) is a key extracellular signaling molecule in neuropathic pain. LPA and its receptors are responsible for long-lasting mechanical allodynia and thermal hyperalgesia as well as demyelination and upregulation of pain-related protein expression patterns in a variety of animal models. Evidence supports the hypothesis that antagonism of the LPA signaling pathways would therefore benefit the treatment of neuropathic pain. Diabetic neuropathy in Type 1 and 2 diabetes in both humans and animal models is characterized by pathophysiological changes in all components (sensory, motor and autonomic) of the peripheral nervous system. In addition to the changes of primary afferent nerves, central sensitization is believed to be an important mechanism underlying persistent pain, including neuropathic and inflammatory pain. The experiments described

here were designed to examine the upregulation of LPA in STZ-induced diabetes which, if positive, would suggest that treatment of diabetic rats with the murine anti-LPA antibody (B3) would ameliorate pain behaviors associated with diabetes. B3 is the murine version of the lead humanized anti-LPA antibody LT3114. In order to explore this possibility, studies were performed in the laboratory of Dr. Nigel Calcutt at UCSD using the well-established rat model of STZ-induced Type 1 (insulin deficient) diabetes. The STZ model uses tactile allodynia as a behavioral index of diabetes-induced neuropathic pain.

3 MATERIALS AND METHODS

3.1 Test Article and Control

For Immunohistochemistry:

Humanized anti-LPA antibody (LT3211) Lot#2012083

Control IgG (LT1023) Lot # LP101210

For efficacy study:

Murine anti-LPA antibody (B3) Lot# HR101201

Control IgG (LT1015) Lot# HR91184

3.2 STZ Model Development

Female Sprague-Dawley rats (Harlan Industries, San Diego CA) weighing 225-250 grams each were maintained at room temperature, between 65 to 82°F with relative humidity between 30 to 70%. The room was illuminated with fluorescent lighting on a daily 12 hour light/dark cycle. All animals were maintained 2/cage with free access to dry food and municipal water. Insulin deficient diabetes was induced following an overnight fast by a single IP injection of STZ (55 mg/kg) dissolved in 0.9% sterile saline. Hyperglycemia was confirmed 4 days later and also prior to behavioral testing in a sample of blood obtained by tail prick using a strip operated reflectance meter. All animals were observed daily and weighed regularly during the study period. Previous studies have shown that allodynia in these diabetic animals occurs after 2- 3 weeks post – STZ treatment. Once allodynia was established six animals were sacrificed and blood samples were taken for MS analysis and dorsal root ganglia collected for immunohistochemical analysis. Blood samples were collected in tubes containing potassium EDTA (K₂EDTA) and centrifuged at approximately 5000 rpm for approximately 10 minutes at 4°C. Plasma was harvested, immediately frozen on dry ice and stored at -70 ± 10°C until assayed.

3.3 Measurement of plasma LPA concentrations

Mass Spectrometry of LPA in plasma was carried out according to the method described by Zhao.¹ In brief, lipids were extracted from plasma samples using a two-step extraction.² Following dilution (1:25) in phosphate buffered saline 0.5 ml was vortexed on ice with 2 ml of MeOH/chloroform (2:1). A further 2 ml of chloroform/water (1:1) was added to separate the phases. The lower phase was removed and a further 1ml of chloroform was added to the upper phase vortexed and the lower phase separated again. Lower phase extracts were combined and solvent evaporated under nitrogen.

The dried lipids were resuspended in 100 ul of methanol for MS analysis (triple quadrupole ESI-MS) and were loaded through and LC system with autosampler. An internal standard was used for quantitation purposes.

3.4 Immunohistochemistry of LPA in Dorsal Root Ganglia

Frozen tissue sections (DRG, 5 µm) were fixed in 10% neutral buffered formalin and antigen retrieved following the ALTER technique overnight. Sections were blocked with 2% casein solution, drained and incubated with primary antibodies against LPA (LT3211, humanized anti-LPA antibody) or IgG control (LT1023, human IgG) at 5 ug/ml in TBS at room temperature for 1 hour. Immunoreactivity was visualized with ABC Elite anti-human system from Vector Lab following the vendor instructions.

3.5 Efficacy study

3.6 Study Design

To investigate the efficacy of B3 in diabetic induced neuropathic pain, three groups of diabetic rats were established by treatment with STZ. Approximately three weeks after the onset of diabetes, animals were dosed both intravenously and intrathecally to ensure antibody access to both the peripheral and central nervous system compartments. One group of animals received 10 mg/kg of B3 administered intravenously concomitantly with 10 ug of B3 via an intrathecal catheter. A second group received 10 mg/kg of IgG control antibody administered intravenously concomitantly with 10 ug of IgG intra-theccally. A third group of diabetic animals received vehicle alone and non-diabetic rats were used as normal controls. The paw withdrawal threshold was measured at the start of the study (baseline) and at 3 hours after dosing.

3.7 Paw Withdrawal Threshold

Rats were transferred to a testing cage with a wire mesh bottom and allowed to acclimate. Von Frey filaments (Stoelting, Wood Dale IL) were used to determine the 50% mechanical threshold for foot withdrawal. A series of filaments, starting with one possessing a buckling weight of 2.0g, were applied in sequence to the plantar surface of the right hindpaw with a

pressure that causes the filament to buckle. Lifting of the paw was recorded as a positive response and the next lightest filament chosen for the next measurement. Absence of a response after 5 seconds prompted use of the next filament of increasing weight. This paradigm was continued until four measurements were made after an initial change in the behavior or until five consecutive negative (given the score of 15g) or four positive (score of 0.25g) scores occurred. The resulting sequence of positive and negative scores was used to interpolate the 50% response threshold. Only rats with a 50% tactile response threshold below 6 g are considered allodynic and brought forward for drug testing.

3.8 Sample Collection and Analysis

Blood samples were collected in tubes containing potassium EDTA (K₂EDTA) 3 hours after the dose of antibody. Blood samples were centrifuged at approximately 5000 rpm for approximately 10 minutes at 4°C, and plasma was harvested. Plasma samples were immediately frozen in dry ice and stored at -70 ± 10°C until assayed.

3.9 Measurement of total LPA concentration

Bioactive LPA was estimated from the ability of LPA present in plasma samples to activate the LPA₁ receptor. The reporter cell assay (DiscoverX) used the CHO-K1 EDG2 (LPA₁ receptor) β -arrestin clone (PathHunter™) that uses Enzyme Fragment Complementation technology in which two weakly complementing fragments of the B-galactosidase (β -gal) enzyme are expressed in stably transfected cells. One fragment of the β -gal, termed the enzyme acceptor (EA), is fused to the C-terminus of β -arrestin2. The complementing fragment of β -gal, termed the ProLink™ tag, is expressed as a C-terminal fusion protein with the LPA₁ receptor. Upon activation, the LPA₁ receptor is phosphorylated, providing a binding site for β -arrestin. The interaction of β -arrestin and the LPA₁ receptor forces the interaction of ProLink and EA, thus allowing complementation of the two fragments of β -gal and the formation of functional enzyme capable of hydrolyzing substrate and generating a chemiluminescent signal. Briefly, CHO-K1 EDG2 cells were plated at and incubated at 37°C/5% CO₂. After 24 hours, plates were starved with reduced serum media at 37°C for another 24 hours. LPA standard curves were prepared by titrating LPA (0 – 60 μ M) in F12 media containing 1mg/mL BSA onto the plate to determine low and maximal signal (dotted lines on Figure 2). Standards and samples were added to the plate and incubated at room temperature for 90 minutes. The plates were washed, incubated with the detection reagent in the dark for 90 minutes and read on a standard luminescence plate reader.

3.10 Estimation of free LPA concentration

The amount of B3 not bound to LPA (free B3) in plasma samples collected from STZ-induced diabetic rats was investigated using the Kinetic Exclusion Assay (KinExA, Sapidyn Instruments, Boise, ID, USA). The LPA conjugate, which is used to capture free B3 in the sample, was prepared by crosslinking 1-(18-mercaptooctadecanoyl)-2-hydroxy-/sn/-glycero-3-phosphate to maleimide-activated BSA (Thermo Scientific, Waltham, MA, USA) in 0.1 M

sodium phosphate, 0.15 M NaCl, pH 7.2. The purified LPA-BSA conjugate was diluted to 30 µg/mL with PBS (without calcium and magnesium (Cellgro, Manassas, VA, USA)), absorbed to PMMA beads (Sapidyne Instruments, Boise, ID, USA), blocked with 10 mg/mL Fraction V fatty acid-free BSA (FAF-BSA, Calbiochem, USA), and re-suspended in running buffer (PBS with 2 mg/mL FAF-BSA, 0.02% TWEEN 20, and 0.002% azide). The standard curves were generated using purified B3 antibody (Lot HR101201) diluted in running buffer.

Using a glass syringe, the standard curve samples (in duplicate) and the plasma samples (in triplicate) were injected directly upstream of the flow cell to minimize the time between sample injection and free B3 capture. In this format, B3-LPA complexes were not allowed to dissociate, so only the free B3 in the neat plasma sample was detected. For the total antibody measurements, the plasma samples were diluted in running buffer to final dilutions of 1:1,000, 1:2000, and 1:5,000 in glass tubes and incubated at room temperature for 12 hours. Under these dilute conditions, the B3-LPA complexes dissociate so nearly all the antibody is unoccupied at equilibrium.

For both the free and total antibody measurements, B3 captured on the beads was detected using a DyLight goat anti-mouse heavy and light chain secondary (Jackson ImmunoResearch, West Grove, PA, USA) at 125 ng/mL in running buffer. Hyperbolic interpolation was carried out using KinExA Pro software (Sapidyne Instruments, Boise, ID, USA) and calculated plasma levels of B3 were averaged. The concentration of B3-LPA complex in the plasma was determined by subtracting the measured free B3 concentration from the measured total B3 concentration. By definition, the ratio of $[B3-LPA]/[B3]_{Free}$ equals the ratio of free LPA to the equilibrium dissociation constant ($[LPA]_{Free}/K_D$).

3.11 Data Analyses

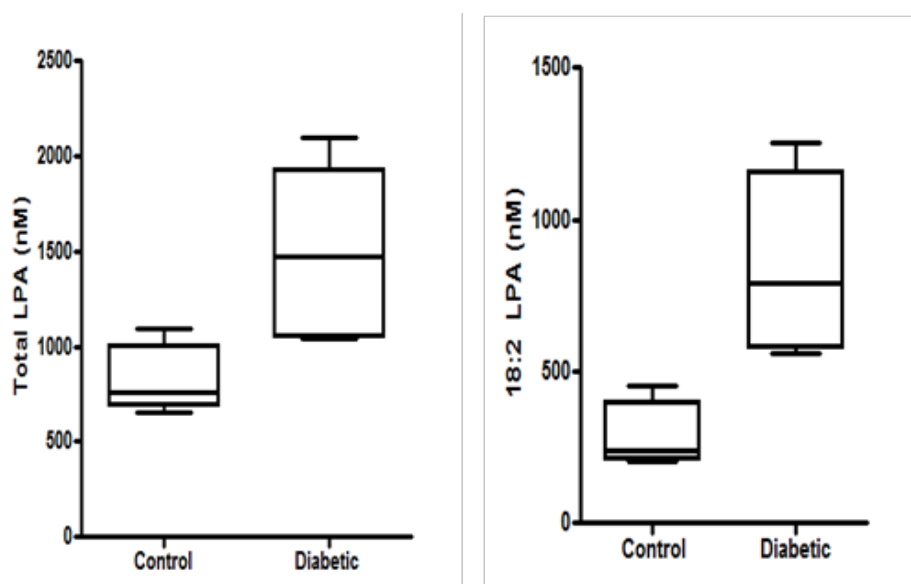
Animal studies were performed at the laboratory of Dr. Nigel Calcutt at UCSD as were measurements of mechanical allodynia. Mass Spectrometry analysis was carried out by Dr. Yan Xu at the Indiana University School of Medicine. All other analyses were performed at Lpath Inc. Descriptive statistics was used to calculate median, mean, standard deviation (SD) and standard error of the mean (SEM) using Microsoft Excel. An excel spread sheet containing all data and analysis was sent via e-mail to Lpath. GraphPad Prism software (version 6.0, Graph Software inc, La Jolla, CA) was used to graph and analyze (ANOVA) the data.

4 RESULTS

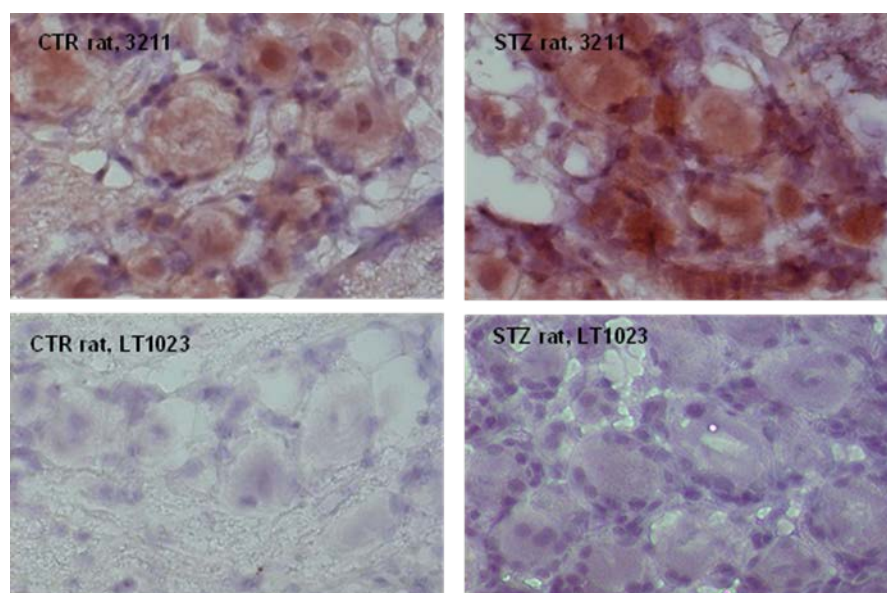
4.1 Model Development

In diabetic rats that were allodynic, levels of LPA were elevated in the plasma (Figure 1) and the dorsal root ganglia (Figure 2) in comparison with normal control rats. Staining in the dorsal root ganglia of diabetic rats was specific to the presence of LPA as shown by the lack of staining in these samples of a non-specific control antibody (Figure 2).

Figure 1: Plasma LPA concentrations in Streptozocin rats



Diabetes induced by a single i.p. injection of streptozocin (55 mg/kg) and samples collected 3 weeks after induction of diabetes. Plasma LPA measured by LC-MS.

Figure 2: Upregulation of LPA in the Dorsal Root Ganglia of STZ rats

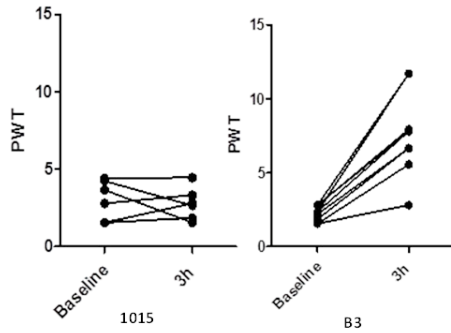
LPA staining in DRG of Control (CTR) and Diabetic (STZ) rats. Frozen DRG sections (5 μ m) were fixed in 10% neutral buffered formalin, antigen retrieved and then stained with an anti-LPA antibody (LT3211) and an Isotype matched IgG1 Ab (LT1023) as control. The ABC Elite system from Vector Lab was used as the detection system for the staining. Strong LPA signal was localized particularly in medium and small ganglion bodies from STZ rats compared with control animals. No signal was obtained for LT1023 staining.

These data show that LPA concentrations are affected by the development of diabetes but are not indicative of whether the elevated concentrations are actually causing the allodynia seen.

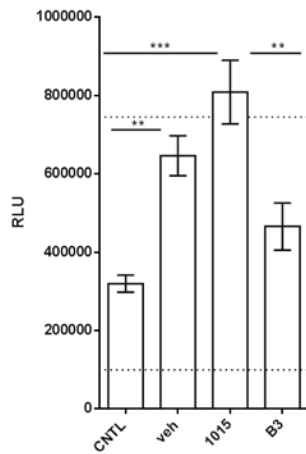
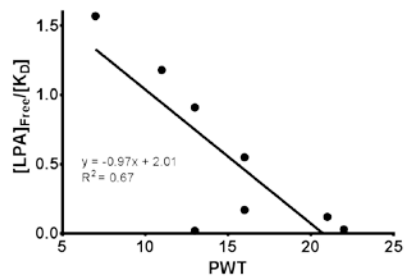
4.2 Efficacy study

A single intravenous injection of B3 (10mg/kg) combined with an intra-theal injection (10 ug) alleviated tactile allodynia when compared with diabetic rats treated with either saline (vehicle group) or isotype control IgG (LT1015); Figure 3A.

The levels of bioactive LPA (defined as the amount of LPA present in plasma able to activate LPAR1 in a cell based assay) measured in these animals demonstrated that there was an increase in bioactive LPA in diabetic animals which was significantly reduced in animals treated with B3 (Figure 3B). Moreover, the data show that there was an inverse correlation between PWT and residual free plasma LPA (Figure 3C).

Figure 3: Efficacy of anti-LPA antibody on LPA levels and established neuropathic pain**A) Paw Withdrawal Threshold (PWT) following treatment with B3**

Tactile responses, represented as 50% PWT, in rats receiving combined intravenous (10mg/kg) and intra-theal (10ug) injection of anti-LPA antibody (B3) or control IgG (1015).

B) Concentration of LPA in plasma of STZ rats**C) Correlation between estimated free [LPA] and PWT**

Plot of [L]/K_D vs. AUC reveals a strong correlation between the ability of the antibody to decrease the free LPA concentration in plasma and the efficacy of the antibody to protect the animal from pain.

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These data demonstrate that blocking the activity of LPA with a single dose of an anti-LPA antibody is sufficient to ameliorate neuropathic pain in the rat STZ-induced model of diabetic neuropathy and that the reduction in pain is accompanied by a decrease in LPA levels that are upregulated in the vehicle group.

5 DATA ARCHIVE

Lpath notebook #137 p. 29; #250 pp. 32-33; #333 pp. 1-10; #209 pp.47-52,54-57,65-69, 73-74.

6 LIST OF REFERENCES

1. Zhao Z and Xu Y. Measurement of endogenous lysophosphatidic acid by ESI-MS/MS in plasma samples requires pre-separation of lysophosphatidyl choline. *J. Chrom. B.* 2009;877:3739-3742.
2. Zhao Z, Xiao Y, Elson P, et al. Plasma lysophosphatidylcholine levels: Potential biomarkers for colorectal cancer. *J. Oncol Res.* 2007;26:2696-2701.